Lecture 2

The box in time

Objectives. *Every box state evolves, including states outside the Hamiltonian domain. The Schrödinger equation holds exactly on that domain. We compute an oscillating position density and the revivals, then apply the interval reading postulate. For finite sums of levels, the walls alone change the mean momentum.*

- Evolve a sine expansion and identify when it is differentiable.
- Compute interval probabilities and the state after a yes.
- Derive the wall-force formula for finite sine sums.

2.1 Every state evolves

In *Quantum Mechanics I*, Thm 4.4, a system of n levels with the Hamiltonian $H = \sum_k \varepsilon_k E_k$, with the projections E_k on its levels, evolves by the unitaries $U_t = \sum_k e^{-\mathrm{i}t\varepsilon_k/\hbar} E_k$. The box has infinitely many levels E_n, each with the eigenvector u_n. Recall that $c_n = \langle u_n, \psi\rangle$.

Definition 2.1 (Evolution in the box)**.** For a real t and $\psi \in L^2(0, L)$, put

$$U_t\psi = \sum_{n=1}^{\infty} e^{-\mathrm{i}E_n t/\hbar}\, c_n u_n. \tag{2.1}$$

The coefficients $e^{-\mathrm{i}E_n t/\hbar} c_n$ have the moduli $|c_n|$, so the series converges by the sine basis theorem (Theorem 1.3). Unlike H, the operator U_t acts on every vector.

Theorem 2.2 (Every state evolves)**.**

1. *Each U_t is unitary, $U_0 = \mathbb{1}$, and $U_{t+s} = U_t U_s$ for all real t and s.*
2. *For every ψ, the map $t \mapsto U_t\psi$ is continuous, in norm.*
3. *The operator U_t maps $D(H)$ onto $D(H)$. If $\psi \in D(H)$, then $t \mapsto U_t\psi$ is differentiable in norm at every t, and*
$$\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}t} U_t\psi = HU_t\psi. \tag{2.2}$$
If $\psi \notin D(H)$, then $t \mapsto U_t\psi$ is differentiable at no t.
4. *For a unit vector ψ, the energy probabilities $|\langle u_n, U_t\psi\rangle|^2 = |c_n|^2$ do not depend on t.*

Proof.

1. *A group of unitaries.* The phases multiply, $e^{-\mathrm{i}E_n(t+s)/\hbar} = e^{-\mathrm{i}E_n t/\hbar} e^{-\mathrm{i}E_n s/\hbar}$, so $U_{t+s} = U_t U_s$, and $U_0 = \mathbb{1}$. By the inner product formula (1.6), $\langle U_t\varphi, U_t\psi\rangle = \langle \varphi, \psi\rangle$, so $U_t^\dagger U_t = \mathbb{1}$. Since also $U_t U_{-t} = \mathbb{1}$, we get $U_t^\dagger = U_{-t}$, and U_t is unitary.

2. *Continuity.* By Parseval's identity,
$$\|U_t\psi - U_s\psi\|^2 = \sum_n \left|e^{-iE_nt/\hbar} - e^{-iE_ns/\hbar}\right|^2 |c_n|^2 .$$
Given $\epsilon > 0$, choose M with $\sum_{n>M} |c_n|^2 < \epsilon$. Each term is at most $4|c_n|^2$, so the terms with $n > M$ add to less than 4ϵ. The first M terms tend to 0 as $s \to t$. So $\|U_t\psi - U_s\psi\|^2 < 5\epsilon$ for s close to t.
3. *The derivative on the domain.* The coefficients of $U_t\psi$ have the moduli $|c_n|$, so $U_t\psi \in D(H)$ exactly when $\psi \in D(H)$, and U_{-t} inverts U_t there. Let $\psi \in D(H)$. Since $U_{t+h}\psi - U_t\psi = U_t(U_h\psi - \psi)$ and U_t is unitary, it suffices to differentiate at $t = 0$. By Parseval's identity,
$$\left\|\frac{U_h\psi - \psi}{h} + \frac{i}{\hbar}H\psi\right\|^2 = \sum_n \left|\frac{e^{-iE_nh/\hbar} - 1}{h} + \frac{iE_n}{\hbar}\right|^2 |c_n|^2 .$$
Each term tends to 0 as $h \to 0$. Since $\left|e^{-i\theta} - 1\right| \le |\theta|$ for real θ, each term is at most $(2E_n/\hbar)^2 |c_n|^2$, and these add to a finite sum because $\psi \in D(H)$. The argument of the continuity step then shows that the sum tends to 0. So the derivative at 0 is $-(i/\hbar)H\psi$, and at t it is $-(i/\hbar)U_tH\psi = -(i/\hbar)HU_t\psi$, which is (2.2).
4. *No derivative outside the domain.* Suppose that $(U_{t+h}\psi - U_t\psi)/h$ tends to a vector χ as $h \to 0$. The inner product with u_n is continuous, so
$$\langle u_n, \chi\rangle = \lim_{h\to 0} \frac{e^{-iE_n(t+h)/\hbar} - e^{-iE_nt/\hbar}}{h} c_n = -\frac{iE_n}{\hbar} e^{-iE_nt/\hbar} c_n .$$
By Parseval's identity, $\sum_n E_n^2 |c_n|^2 = \hbar^2 \|\chi\|^2 < \infty$, so $\psi \in D(H)$.
5. *Energy.* The phase $e^{-iE_nt/\hbar}$ has modulus 1. ∎

So every state of the box evolves, and the Schrödinger equation holds exactly for the states in the domain of H. The state of § 1.7, with $c_n \propto 1/n^2$, evolves continuously, and at no time does it have a time derivative. Here the continuum changes the answer about evolution. The equation of motion holds on the domain, and the evolution acts on every state. The same holds for every self-adjoint Hamiltonian (Prugovečki, Thm IV.3.1; Amrein, Prop. 5.1), and Lecture 7 returns to it.

Self-check 2.1

1. For which ψ does $U_t\psi$ represent the same state as ψ for every t?
2. Let $c_n = C\,n^{-r}$ with $r > 0$ and a constant $C \neq 0$. For which r do the c_n define a state? A state of finite mean energy? A state whose evolution has a time derivative? Which case is the parabola of Example 1.9?

2.2 Two levels beat

A stationary wave has a density that does not move. The evolved state $U_tu_n = e^{-iE_nt/\hbar}u_n$ differs from u_n by a phase, so its density $|(U_tu_n)(x)|^2 = u_n(x)^2$ is constant in time. A superposition of two levels does move.

Example 2.3 (Two levels beat).

1. *Prepare and evolve.* Take $\psi = (u_1 + u_2)/\sqrt{2}$. By (2.1),
$$\psi_t = U_t\psi = \frac{1}{\sqrt{2}}\left(e^{-iE_1t/\hbar}u_1 + e^{-iE_2t/\hbar}u_2\right).$$

2. *Find the density.* The u_n are real and $E_2 - E_1 = 3E_1$, so
$$|\psi_t(x)|^2 = \frac{1}{2}\big(u_1(x)^2 + u_2(x)^2\big) + u_1(x)u_2(x)\cos\frac{3E_1 t}{\hbar}.$$
3. *Weigh the left half.* Since $\int_0^{L/2} u_1^2\,\mathrm{d}x = \int_0^{L/2} u_2^2\,\mathrm{d}x = \frac{1}{2}$ and $\int_0^{L/2} u_1 u_2\,\mathrm{d}x = 4/3\pi$,
$$\int_0^{L/2} |\psi_t(x)|^2\,\mathrm{d}x = \frac{1}{2} + \frac{4}{3\pi}\cos\frac{3E_1 t}{\hbar}.$$
The weight swings between $\frac{1}{2} - 4/3\pi \approx 0.076$ and $\frac{1}{2} + 4/3\pi \approx 0.924$, with the period $2\pi\hbar/3E_1$.
4. *Read the energy.* By Theorem 2.2, the energy reads E_1 or E_2, each with probability $\frac{1}{2}$, at every time.

§ 2.4 makes this weight the probability of finding the particle in the left half. The density of the state moves, while its energy probabilities do not. The density oscillates at the frequency $(E_2 - E_1)/\hbar$, a Bohr frequency of *Quantum Mechanics I*, Lecture 4. Figure 2.1 plots the weight of the left half over three periods.

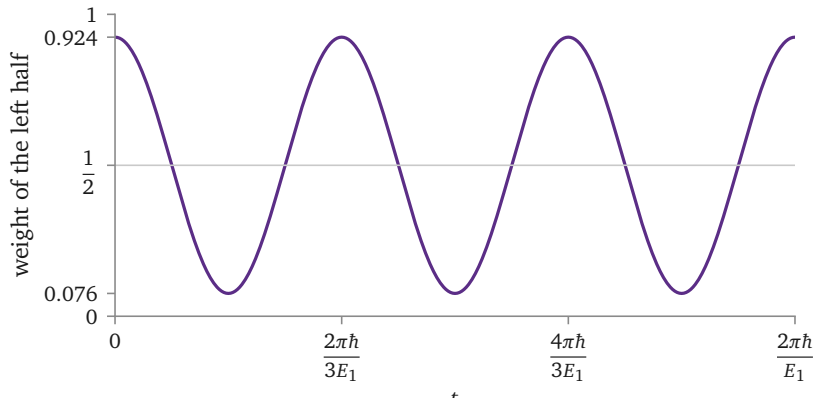


Figure 2.1: The weight of the left half for the state $\psi = (u_1 + u_2)/\sqrt{2}$ of Example 2.3, which is $\frac{1}{2} + (4/3\pi)\cos(3E_1 t/\hbar)$, over three periods $2\pi\hbar/3E_1$. The thin line marks the value $\frac{1}{2}$.

Self-check 2.2

1. Starting from $(u_1 + \mathrm{i}u_2)/\sqrt{2}$ at $t = 0$, find the weight of the left half at the time t.

2.3 Revivals

The energies $E_n = n^2 E_1$ are integer multiples of E_1, so all the phases in (2.1) return together.

Proposition 2.4 (Revivals). *Let $T = 4mL^2/\pi\hbar$. Then $U_T = \mathbb{1}$, and for every $\psi \in L^2(0, L)$,*
$$(U_{T/2}\psi)(x) = -\psi(L - x), \tag{2.3}$$
as elements of $L^2(0, L)$.

Proof. First, $E_n T/\hbar = 2\pi n^2$, so every phase $e^{-\mathrm{i}E_n T/\hbar}$ is 1, and $U_T\psi = \psi$. Next, $E_n T/2\hbar = \pi n^2$, and $e^{-\mathrm{i}\pi n^2} = (-1)^n$, since n^2 and n have the same parity. The reflection $(R\psi)(x) = \psi(L-x)$ preserves norms and satisfies $R^2 = \mathbb{1}$, so it is unitary, as in Theorem 2.2, and $Ru_n = (-1)^{n+1}u_n$ because $\sin(n\pi - y) = (-1)^{n+1}\sin y$. A unitary operator may be applied to a norm-convergent series term by term. Hence

$$U_{T/2}\psi = \sum_n (-1)^n c_n u_n = -\sum_n c_n\, Ru_n = -R\psi. \qquad \blacksquare$$

The reflection R is the *parity* of the box. It multiplies each coefficient c_n by $(-1)^{n+1}$, which keeps $|c_n|$. So R maps $D(H)$ onto itself, and $RH\psi = HR\psi$ for every $\psi \in D(H)$. Parity is why the even levels were missing from the parabola of Example 1.9.

Since $E_1 T/\hbar = 2\pi$, the revival time is $T = 2\pi\hbar/E_1$. The beats of Example 2.3 have the period $2\pi\hbar/3E_1 = T/3$. Every state of the box returns exactly after the time T, and at $T/2$ it is the mirror image of itself, up to a sign that no reading detects. Classical particles of different energies bounce with different periods, and nothing brings them all back at once. The quantum revival comes from the energies $n^2 E_1$.

Over short times a packet does behave like a classical particle. Take the Gaussian packet

$$\psi_0(x) = C\exp\Big(-\frac{(x-0.35L)^2}{4(0.1L)^2}\Big)e^{30\pi \mathrm{i}x/L} \qquad (0 < x < L),$$

with the constant $C > 0$ fixed by $\|\psi_0\| = 1$. Its coefficients $|c_n|$ are largest at $n_0 = 30$, and it has the central momentum $n_0\pi\hbar/L$. Figure 2.2 shows its density, computed from (2.1) with the series cut at $n = 80$, which keeps more than $1 - 10^{-4}$ of the norm. Its speed $n_0\pi\hbar/mL$ takes it across the box and back in the time $2mL^2/n_0\pi\hbar = T/2n_0$, and for the first few bounces it follows the classical particle. The levels are not equally spaced, so the packet spreads and soon fills the box. Much later, at $T/2$, it is its mirror image, up to a sign.

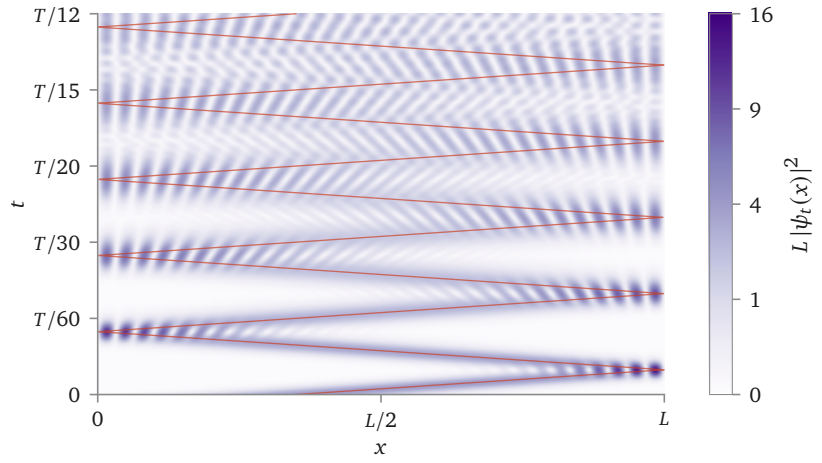


Figure 2.2: The density $|\psi_t(x)|^2$ of the packet ψ_0, for $0 \le t \le T/12$. Darker is denser, and the key gives $L\,|\psi_t(x)|^2$. The colour scale is the square root of the density, so that the spread packet stays visible. The thin line is a classical particle with the packet's central momentum $n_0\pi\hbar/L$, which starts at $x = 0.35L$ moving right and bounces with the period $T/2n_0 = T/60$.

Between 0 and T there are fractional revivals. At $T/4$ the phase $e^{-\mathrm{i}\pi n^2/2}$ is 1 for even n and $-\mathrm{i}$ for odd n. The parts of ψ along the u_n with odd and even n are $(\psi + R\psi)/2$ and

$(\psi - R\psi)/2$, so

$$U_{T/4}\psi = \frac{\psi - R\psi}{2} - \mathrm{i}\,\frac{\psi + R\psi}{2}.$$

A bump in the left half becomes the bump and its mirror image, each with half the weight. Figure 2.3 shows this for

$$\psi(x) = C\,(x - L/8)^2(3L/8 - x)^2 \quad \text{on } [L/8, 3L/8],$$

with $\psi = 0$ elsewhere and the constant $C > 0$ fixed by $\|\psi\| = 1$. By the same formula, at $T/4$ the packet ψ_0 is a superposition of itself and its mirror image.

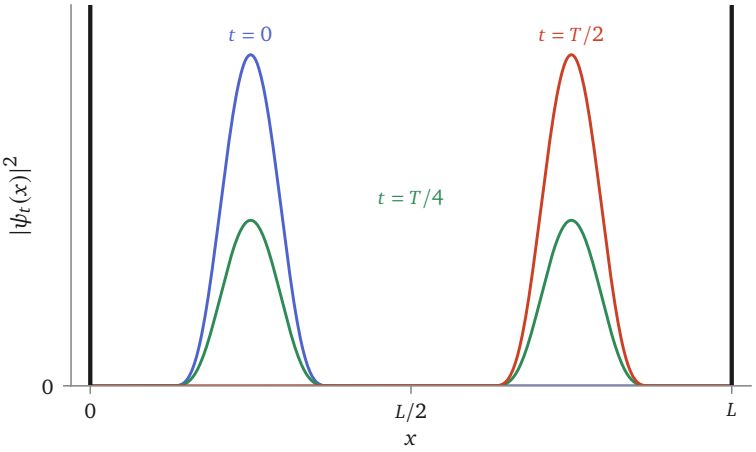


Figure 2.3: The density $|\psi_t(x)|^2$ of the bump, at $t = 0$, at $t = T/4$ (the bump and its mirror image, each with half the weight) and at $t = T/2$ (the mirror image). The curves are computed from (2.1) with the series cut at $n = 120$.

Self-check 2.3

1. What is the weight of the left half for the state of Example 2.3 at $t = T/2$? Check it against (2.3).
2. For $\psi = (u_1 + u_2 + u_3)/\sqrt{3}$, find the first time $t > 0$ at which $U_t\psi$ represents the same state as ψ, and the first at which $U_t\psi = \psi$. Compare with the state of Example 2.3.

The physics

Revivals have been seen in atoms. A short laser pulse puts an electron of a potassium atom into a superposition of highly excited levels, a wave packet that moves in and out, towards the nucleus and away from it. Its levels are not those of a box. The packet spreads out and later reassembles, and Yeazell, Mallalieu and Stroud observed this collapse and revival in 1990. In the box, the density $|\psi_t(x)|^2$ drawn over x and t up to the time T shows regular patterns of lines, now called quantum carpets. Figure 2.2 shows the first twelfth of one. Berry studied their fractal structure in 1996.

2.4 Reading position on an interval

On the chain of § 1.1, the postulates of *Quantum Mechanics I* give the probability of finding the particle on a set of sites. The reading "the particle is on one of the sites in the set S"

has the projection $\sum_{j\in S} |e_j\rangle\langle e_j|$ and the probability $\sum_{j\in S} |\psi_j|^2$. In the box there are no sites, and an element of $L^2(0,L)$ has no value at a point (§ 1.2). So the chain's rule $|\psi_j|^2$ has no analogue at a point. We read position through the question "is the particle in the interval I?".

Definition 2.5 (The projection of an interval)**.** For an interval $I \subseteq [0,L]$, the operator 1_I keeps a wavefunction on I and sets it to zero elsewhere:

$$(1_I\psi)(x) = \begin{cases} \psi(x) & \text{for } x \in I, \\ 0 & \text{for } x \notin I. \end{cases}$$

The operator 1_I acts on every vector, and $1_I 1_I = 1_I$. For all φ and ψ,

$$\langle \varphi, 1_I\psi\rangle = \int_I \overline{\varphi(x)}\,\psi(x)\,\mathrm{d}x = \langle 1_I\varphi, \psi\rangle\,.$$

So 1_I is an orthogonal projection, as in *Quantum Mechanics I*.

Postulate 2.6 (Reading position on an interval)**.** Let ψ be a unit vector. An ideal reading of "the particle is in I" on the state of ψ answers yes with the probability

$$\mathrm{Prob}(I) = \|1_I\psi\|^2 = \int_I |\psi(x)|^2\,\mathrm{d}x. \tag{2.4}$$

An answer that occurs has positive probability. If it is yes, the state immediately afterwards is the ray of $1_I\psi/\|1_I\psi\|$. If it is no, the state is the ray of $(\mathbb{1}-1_I)\psi/\|(\mathbb{1}-1_I)\psi\|$.

This is Born's and Lüders' rule of *Quantum Mechanics I*, Postulate 4.1, for the projection 1_I, the analogue of the chain's projections on sets of sites. The chain motivates the postulate. It does not derive it. By (2.4), $|\psi(x)|^2$ is a probability density, and the weight of Example 2.3 is the probability of finding the particle in the left half. A single point is an interval of length zero, so the answer "at x" has probability 0. A reading with sharp edges is an idealisation, and real detectors read with blurred edges. *Quantum Mechanics III: Foundations* describes such readings, and Lecture 5 puts the interval rule into the general rule for every observable.

Self-check 2.4

1. Find the probability of finding the particle of Example 1.9 in $[0, L/4]$.

2.5 Means in time

The mean position describes the motion of the density. By (2.4), $|\psi_t(x)|^2$ is the probability density of position at the time t, and its mean

$$\bar{x}(t) = \int_0^L x\,|\psi_t(x)|^2\,\mathrm{d}x$$

is the *mean position*. For the state of Example 2.3, the densities u_1^2 and u_2^2 are symmetric about $L/2$, so each contributes $L/2$, and the cross term contributes $\int_0^L x\,u_1u_2\,\mathrm{d}x = -16L/9\pi^2$ (the self-check of § 1.6). So

$$\bar{x}(t) = \frac{L}{2} - \frac{16L}{9\pi^2}\cos\frac{3E_1 t}{\hbar}.$$

The mean swings between $0.320L$ and $0.680L$, with the period $T/3$ of the beats. It turns well before the walls. The mean position is not the path of a particle.

Schrödinger's rule $-(\hbar^2/2m)\,\psi''$ is $p^2/2m$, with p the rule $-\mathrm{i}\hbar\,\mathrm{d}/\mathrm{d}x$ applied twice. We call

$$\bar{p}(t) = \langle \psi_t, -\mathrm{i}\hbar\psi_t' \rangle$$

the *mean momentum*. Which operator is the momentum in the box is Lecture 3's question.

Proposition 2.7 (Means in time)**.** *Let ψ be a unit vector that is a finite combination of the u_n, and let $\psi_t = U_t\psi$. Then the mean momentum $\bar{p}(t)$ is real, and*

$$m\,\frac{\mathrm{d}\bar{x}}{\mathrm{d}t} = \bar{p}, \qquad \frac{\mathrm{d}\bar{p}}{\mathrm{d}t} = \frac{\hbar^2}{2m}\big(|\psi_t'(0)|^2 - |\psi_t'(L)|^2\big).$$

Proof. We differentiate each mean under the integral, use the equation of motion, and integrate by parts. The derivative of $\bar{x}$ keeps an integral over the box, which is $\bar{p}/m$. The derivative of $\bar{p}$ reduces to terms at the walls.

Write ψ for ψ_t. As a finite sum, $\psi_t(x) = \sum_n e^{-\mathrm{i}E_nt/\hbar}c_nu_n(x)$ is smooth in t and x and vanishes at the walls. Since $-(\hbar^2/2m)\,u_n'' = E_nu_n$,

$$\partial_t\psi = \frac{\mathrm{i}\hbar}{2m}\,\psi''.$$

First, the real part of $\overline{\psi}\psi'$ is $\frac{1}{2}(|\psi|^2)'$, whose integral is $\frac{1}{2}(|\psi(L)|^2 - |\psi(0)|^2) = 0$. So $\bar{p} = \hbar\int_0^L \mathrm{Im}(\overline{\psi}\psi')\,\mathrm{d}x$, which is real. Next, $\partial_t\,|\psi|^2 = 2\,\mathrm{Re}(\overline{\psi}\,\partial_t\psi) = -(\hbar/m)\,\mathrm{Im}(\overline{\psi}\psi'')$, and $\mathrm{Im}(\overline{\psi}\psi'') = (\mathrm{Im}(\overline{\psi}\psi'))'$, since $\overline{\psi'}\psi'$ is real. Integrating by parts, with the boundary term $x\,\mathrm{Im}(\overline{\psi}\psi')$ zero at both walls,

$$\frac{\mathrm{d}\bar{x}}{\mathrm{d}t} = -\frac{\hbar}{m}\int_0^L x\,\big(\mathrm{Im}(\overline{\psi}\psi')\big)'\,\mathrm{d}x = \frac{\hbar}{m}\int_0^L \mathrm{Im}(\overline{\psi}\psi')\,\mathrm{d}x = \frac{\bar{p}}{m}.$$

Finally, $\partial_t(\overline{\psi}\psi') = (\mathrm{i}\hbar/2m)(\overline{\psi}\psi''' - \overline{\psi''}\psi')$, so $\mathrm{d}\bar{p}/\mathrm{d}t = (\hbar^2/2m)\int_0^L \mathrm{Re}(\overline{\psi}\psi''' - \overline{\psi''}\psi')\,\mathrm{d}x$. Integrating $\overline{\psi}\psi'''$ by parts, with $\psi = 0$ at the walls, turns it into $-\overline{\psi'}\psi''$, and $\mathrm{Re}(\overline{\psi'}\psi'' + \overline{\psi''}\psi') = (|\psi'|^2)'$. So

$$\frac{\mathrm{d}\bar{p}}{\mathrm{d}t} = -\frac{\hbar^2}{2m}\int_0^L \big(|\psi'|^2\big)'\,\mathrm{d}x = \frac{\hbar^2}{2m}\big(|\psi'(0)|^2 - |\psi'(L)|^2\big). \qquad \blacksquare$$

So $\bar{p}$ is m times the velocity of the mean position, which is why $-\mathrm{i}\hbar\,\mathrm{d}/\mathrm{d}x$ is called the momentum. No force acts inside the box, and the mean momentum changes only at the walls. The left wall raises it at the rate $(\hbar^2/2m)\,|\psi_t'(0)|^2$, and the right wall lowers it at the rate $(\hbar^2/2m)\,|\psi_t'(L)|^2$. For the state of Example 2.3,

$$\bar{p}(t) = m\,\frac{\mathrm{d}\bar{x}}{\mathrm{d}t} = \frac{8\hbar}{3L}\,\sin\frac{3E_1t}{\hbar}.$$

In a stationary wave u_n the two pushes balance, and $\bar{p} = 0$. Each wall alone receives

$$\frac{\hbar^2}{2m}\,u_n'(L)^2 = \frac{2E_n}{L} = -\frac{\mathrm{d}E_n}{\mathrm{d}L},$$

the force with which the particle presses on the wall. It is the rate at which the energy of the level falls as the box widens, as for a gas that pushes a piston.

The proof of Proposition 2.7 also shows how probability moves inside the box. For a finite combination of the u_n, the *probability current* is

$$j(x,t) = \frac{\hbar}{m}\,\mathrm{Im}\big(\overline{\psi_t(x)}\psi_t'(x)\big). \tag{2.5}$$

That proof gives $\partial_t |\psi_t|^2 = -\partial_x j$. Integrating over an interval $[a, b] \subseteq [0, L]$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_a^b |\psi_t(x)|^2 \, \mathrm{d}x = j(a,t) - j(b,t).$$

So $j(x,t)$ is the rate at which probability crosses the point x from left to right. At the walls $\psi_t = 0$, so $j(0,t) = j(L,t) = 0$. For the state of Example 2.3, $j(L/2,t) = (2\pi\hbar/mL^2)\sin(3E_1 t/\hbar)$, and $-j(L/2,t)$ is the time derivative of the weight in Example 2.3. The formula for $\bar{p}$ in that proof says that $\bar{p} = m\int_0^L j(x,t)\,\mathrm{d}x$.

Figure 2.4 shows the density of the state of Example 2.3 at $t = 0$, $T/12$ and $T/6$. Between $t = 0$ and $t = T/6$ the current $j(L/2,t)$ is positive, and probability flows from the left half into the right half. At $t = 0$ the weight of the left half is largest, at $t = T/6$ it is smallest, and at both times the current at $L/2$ is zero.

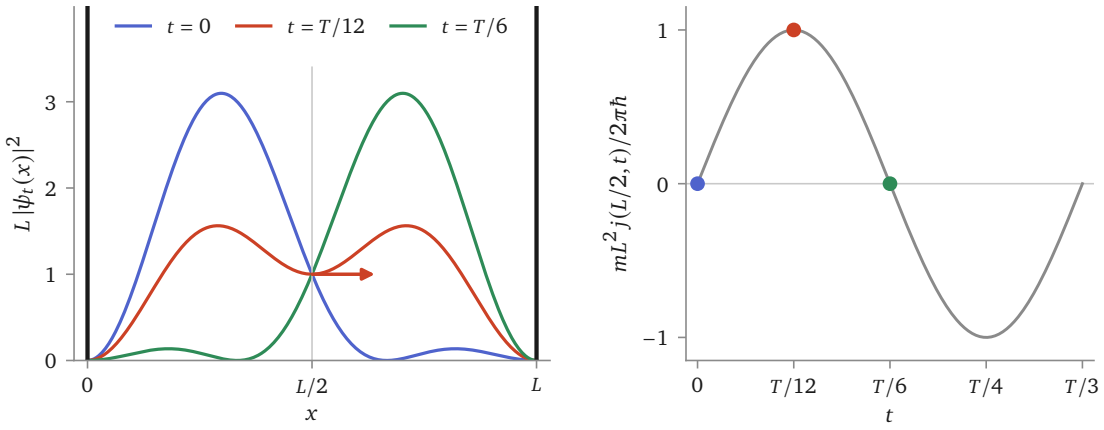


Figure 2.4: Left: the density $L\,|\psi_t(x)|^2$ of the state of Example 2.3 at $t = 0$, $T/12$ and $T/6$. The arrow at $x = L/2$ shows the direction of the current at $t = T/12$. At $L/2$ the density is $1/L$ at every time, while the current there changes sign. Right: the current $j(L/2,t)$ at the midpoint, in units of $2\pi\hbar/mL^2$, over one period $T/3$ of the beats, with the three times marked in the colours of the densities.

Every self-adjoint momentum that Lecture 3 finds acts by $-i\hbar\psi'$ on the finite combinations of the u_n, so they all give the mean $\bar{p}$. Whether they give the same readings is Lecture 3's question.

Self-check 2.5

1. With what force does an electron in the ground state of a box of length $L = 1$ nm press on each wall?
2. For the state of Example 2.3, at which times is $\bar{x}$ least? What is $\bar{p}$ then?

2.6 What a reading leaves

Read "the particle is in the left half" on the ground state u_1. By (2.4) the answer is yes with probability $\frac{1}{2}$. If it is yes, the state afterwards is

$$\varphi = \sqrt{2}\,1_{[0,L/2]}u_1.$$

It jumps at $x = L/2$, from $2/\sqrt{L}$ to 0. Its coefficients are

$$\langle u_n, \varphi \rangle = \frac{2\sqrt{2}}{L} \int_0^{L/2} \sin \frac{n\pi x}{L} \sin \frac{\pi x}{L} \, \mathrm{d}x.$$

For $n = 1$ the integral is $L/4$, so $\langle u_1, \varphi \rangle = 1/\sqrt{2}$. For $n \geq 2$, write the product of sines as $\frac{1}{2}\big(\cos((n-1)\pi x/L) - \cos((n+1)\pi x/L)\big)$ and integrate:

$$\langle u_n, \varphi \rangle = \frac{\sqrt{2}}{\pi}\Big(\frac{\sin((n-1)\pi/2)}{n-1} - \frac{\sin((n+1)\pi/2)}{n+1}\Big).$$

For odd n both sines vanish. For even n they are $-(-1)^{n/2}$ and $(-1)^{n/2}$, so

$$\langle u_n, \varphi \rangle = (-1)^{n/2+1} \, \frac{2\sqrt{2}\, n}{\pi(n^2-1)} \qquad \text{for even } n.$$

A later energy reading gives E_1 with probability $\frac{1}{2}$ and E_2 with probability $32/9\pi^2 \approx 0.360$. Along the even n, $n^2 \, |\langle u_n, \varphi \rangle|^2 \to 8/\pi^2$, so the terms of the mean energy $\sum_n n^2 E_1 \, |\langle u_n, \varphi \rangle|^2$ do not tend to 0, and it is infinite. Before the reading the energy was exactly E_1. We write $\Delta A = \infty$ when the readings of an observable A have an infinite mean square. For the energy, by § 1.7, these are the states outside $D(H)$, and here $\Delta H = \infty$. The state φ still evolves, by Theorem 2.2, without a time derivative.

A finer reading, forgotten, is not the coarser reading. Take $\psi = (u_1 + u_2)/\sqrt{2}$ at $t = 0$ and the left half $I = [0, L/2]$, split into $I_1 = [0, L/4]$ and $I_2 = [L/4, L/2]$. Compare two readings, each followed by a reading of the energy.

- *Read "in I".* The answer is yes with probability $\frac{1}{2} + 4/3\pi$, and the state becomes $1_I\psi/\|1_I\psi\|$. Then E_1 has probability $|\langle u_1, 1_I\psi\rangle|^2 / \|1_I\psi\|^2 = \frac{1}{4} + 2/3\pi \approx 0.462$.
- *Read "in I_1"; if the answer is no, read "in I_2" at once; record only that one answer was yes.* By Postulate 2.6, applied twice, the state is then $1_{I_k}\psi/\big\|1_{I_k}\psi\big\|$ with probability $\big\|1_{I_k}\psi\big\|^2 / \|1_I\psi\|^2$, for $k = 1, 2$. These weights are 0.347 and 0.653, and in the two branches E_1 has probability 0.091 and 0.387. Adding the two branches, E_1 has probability

$$\frac{\big|\langle u_1, 1_{I_1}\psi\rangle\big|^2 + \big|\langle u_1, 1_{I_2}\psi\rangle\big|^2}{\|1_I\psi\|^2} \approx 0.284.$$

Figure 2.5 draws the two readings.

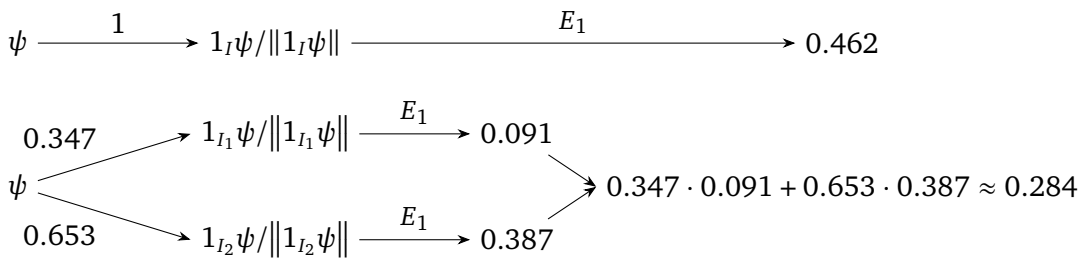


Figure 2.5: The two readings of $\psi = (u_1 + u_2)/\sqrt{2}$, given a yes in I. A yes in I has the probability $\frac{1}{2} + 4/3\pi \approx 0.924$ for both readings. An arrow from ψ carries the probability of the state it reaches. An arrow marked E_1 ends at the probability of E_1 in that state. Upper row: the reading of I leaves one state. Lower rows: the finer reading leaves one of two states, and their contributions add without the cross term.

The numbers come from integrals of products of u_1 and u_2 over the three intervals. For example, $\|1_{I_1}\psi\|^2 = \int_0^{L/4} \frac{1}{2}(u_1+u_2)^2\,\mathrm{d}x = (3\pi - 3 + 4\sqrt{2})/12\pi$. With J one of the intervals:

	$J = I_1$	$J = I_2$	$J = I$
$\|1_J\psi\|^2$	0.320	0.604	0.924
$\langle u_1, 1_J\psi\rangle$	0.170	0.483	0.654
$\|\langle u_1, 1_J\psi\rangle\|^2 / \|1_J\psi\|^2$	0.091	0.387	0.462

Since $1_I = 1_{I_1} + 1_{I_2}$, the two answers differ by the cross term $2\,\langle u_1, 1_{I_1}\psi\rangle\,\langle u_1, 1_{I_2}\psi\rangle\,/\|1_I\psi\|^2$, an interference between the two halves of I. Forgetting which half was found does not undo the finer reading, as for a degenerate reading in *Quantum Mechanics I*, Prop. 4.6.

Self-check 2.6

1. Given the answer yes on u_1, what is the probability that an immediate second reading of the left half answers yes?
2. Given the same yes, the state φ evolves for the time $T/2$. What is then the probability of the left half?
3. Compare the two readings of this section for the state $(u_1 - u_2)/\sqrt{2}$ at $t = 0$, with the same I, I_1 and I_2. Given a yes, find the probability of E_1 for each reading, and explain why the order is the opposite of the order for $(u_1 + u_2)/\sqrt{2}$.

The particle in a box, in time

Evolution: every state evolves by the unitary group $U_t\psi = \sum_n e^{-\mathrm{i}E_n t/\hbar} c_n u_n$. The Schrödinger equation $\mathrm{i}\hbar\,\mathrm{d}\psi_t/\mathrm{d}t = H\psi_t$ holds exactly when $\psi \in D(H)$, and every state returns after the time $T = 4mL^2/\pi\hbar$ (Theorem 2.2 and Proposition 2.4). *Measurement of position:* an ideal reading of "in the interval I" answers yes with probability $\int_I |\psi|^2\,\mathrm{d}x$, and a yes leaves the ray of $1_I\psi/\|1_I\psi\|$ (Postulate 2.6). *Means:* for finite sine sums, the mean position moves with the mean momentum, and only the walls change the mean momentum (Proposition 2.7).

Summary.

- *Every state evolves*: U_t is a unitary group, and $U_t\psi$ has a time derivative, given by the Schrödinger equation, exactly when $\psi \in D(H)$ (Theorem 2.2).
- *Revivals*: every state returns after $T = 4mL^2/\pi\hbar$ and is mirrored at $T/2$ (Proposition 2.4).
- *Reading position on an interval*: probability $\|1_I\psi\|^2$, state after a yes $1_I\psi/\|1_I\psi\|$ (Postulate 2.6). A sharp reading can leave a state of infinite mean energy.
- *Means in time*: for finite sine sums, $m\,\mathrm{d}\bar{x}/\mathrm{d}t = \bar{p}$, and the walls alone change $\bar{p}$. The level u_n presses on each wall with the force $2E_n/L$ (Proposition 2.7).

The mean momentum is the mean of the rule $-\mathrm{i}\hbar\,\mathrm{d}/\mathrm{d}x$. Lecture 3 asks which operator is the momentum of a particle in the box, and what its readings are.

Exercises. Exercise Sheet 2.

Self-check answers.

2.1 1. Exactly the nonzero multiples of a single u_n. If c_m and c_n are both nonzero, with $m \neq n$, the relative phase $e^{-\mathrm{i}(E_m - E_n)t/\hbar}$ changes with t. 2. A state for $r > 1/2$ ($\sum_n n^{-2r} < \infty$), finite mean energy for $r > 3/2$ ($\sum_n n^{2-2r}$), a time derivative for $r > 5/2$ ($\sum_n n^{4-2r}$, the domain of H). The parabola has $c_n \propto n^{-3}$ on the odd n, so its evolution has a time derivative.

2.2 1. $\frac{1}{2} + (4/3\pi)\sin(3E_1t/\hbar)$, since the cross term is $\mathrm{Re}(ie^{-3iE_1t/\hbar}) = \sin(3E_1t/\hbar)$.

2.3 1. $\frac{1}{2} - 4/3\pi \approx 0.076$, since $\cos 3\pi = -1$. By (2.3) it is the weight of the right half at $t = 0$, $1 - 0.924$. 2. Both are T. The relative phases $3E_1t/\hbar$ and $8E_1t/\hbar$ are multiples of 2π only when t is a multiple of $T/3$ and of $T/8$, that is, of T, and then $E_1t/\hbar$ is a multiple of 2π too. The state of Example 2.3 returns as a state at $T/3$, but $U_{T/3}\psi = e^{-2\pi i/3}\psi$, and $U_t\psi = \psi$ first at T.

2.4 1. $30\int_0^{1/4} y^2(1-y)^2\,dy = 53/512 \approx 0.104$.

2.5 1. $2E_1/L \approx 1.2{\times}10^{-10}$ N, with $E_1 \approx 0.376$ eV (the self-check of § 1.3). 2. At $t = 0, T/3, 2T/3, \dots$, where $\bar{x} = L/2 - 16L/9\pi^2 \approx 0.320L$ and $\bar{p} = 0$.

2.6 1. 1, since $1_{[0,L/2]}\varphi = \varphi$. 2. 0: by (2.3), $U_{T/2}\varphi = -R\varphi$ vanishes on the left half. 3. Here $\|1_I\psi\|^2 = \frac{1}{2} - 4/3\pi \approx 0.076$, $\langle u_1, 1_{I_1}\psi\rangle \approx -0.042$ and $\langle u_1, 1_{I_2}\psi\rangle \approx 0.095$. The reading of I keeps the sum of the two amplitudes, and E_1 has the probability $\frac{1}{4} - 2/3\pi \approx 0.038$. The finer reading adds their squares, and E_1 has the probability ≈ 0.143. The cross term is negative here, and it was positive for $(u_1 + u_2)/\sqrt{2}$. The finer reading has no cross term in either case.

References

Prugovečki, *Quantum Mechanics in Hilbert Space*, Thm IV.3.1 (p. 288; his group is e^{iAt}). Amrein, *Hilbert Space Methods in Quantum Mechanics*, Prop. 5.1 (p. 194). *Quantum Mechanics I*, Thm 4.4, Postulate 4.1, Prop. 4.6, Lecture 4. Yeazell, Mallalieu and Stroud, *Physical Review Letters* 64, 2007 (1990). Berry, *Journal of Physics A* 29, 6617 (1996).